

UTM

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PROJECT PROPOSAL

AUTISM EARLY SCREENING SUPPORT SYSTEM

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1.0 Project Background

Nowadays, the current system relies heavily on manual processes and traditional methods. The current autism early screening support process is primarily conducted through healthcare professionals, educators or autism specialists. Parents or guardians who suspect developmental delays in their children are required to visit a healthcare centre, clinic or autism intervention centre for an initial assessment.

During the screening session, parents are required to complete paper-based questionnaires which are known as M-CHAT-R regarding the child's communication skills, social interaction, repetitive behaviours, and developmental milestones. Healthcare professionals then observe the child's behaviour and review the completed questionnaire together with the parents.

Based on the collected information, the healthcare professional performs a preliminary screening to determine whether the child shows signs associated with Autism Spectrum Disorder (ASD). If the screening indicates possible autism traits, the child is referred for a comprehensive diagnostic assessment conducted by qualified specialists. If the child is formally diagnosed with Autism Spectrum Disorder (ASD) by a qualified specialist, the parents or guardians may register the child with the Department of Social Welfare (Jabatan Kebajikan Masyarakat, JKM). To complete the registration, they are required to submit the BPPOKU form together with the necessary supporting documents, such as the medical diagnosis report and identification documents. Upon successful registration, the child may become eligible for welfare assistance, support services, and benefits provided by JKM.

The current process has several limitations. The screening process is time-consuming, relies heavily on manual documentation, and requires parents to attend physical appointments. In addition, there is limited accessibility to specialists in certain areas, and parents often experience delays before receiving screening results or recommendations for further assessment.

2.0 Project Objectives

The recommended solution is known as the Autism Early Screening Support System, which is an advanced two-fold application Web and Mobile technology system. The system combines standard medical screening procedures with daily management systems to create an efficient technological platform for the use of parents, children, and NGOs.

2.1 Core System Infrastructure and Security

The basis of the system lies in data security and privacy.

2.1.1 Authentication and Access Controls

The system uses safe methods for logging into and out of the system, which ensures that confidential data is protected by not allowing access to people who are not authorized to see the data.

2.1.2 Registration and Account Management

A separate module of the system is used to manage users' accounts.

2.1.3 Data Security and Privacy

Considering the confidentiality of data in question, data security is a critical feature of the core infrastructure of the proposed solution, as it will serve as a reliable environment for the formative years intervention program.

2.2 Standardized Screening & Profile Management

This module digitalizes the old screening processes manually performed to guarantee accuracy and fast intervention.

2.2.1 Digital Profile Management (BPPOKU Pindaan 2019)

The program will digitalize Bahagian A (Biodata) from the BPPOKU (Pindaan 2019) form. This process will include the following important pieces of information:

- ➢ Personal Details

  Name, MyKad / MyKid identification, and contact details.

- ➢ Education & Professional Experience

  Education level and parents/guardians' employment.

- ➢ Details of Next of Kin

  The system collects detailed information about relatives close to the person for further communication in case of necessity.

2.2.2 M-CHAT-R Screening Integration

This feature includes a test with 20 questions performed according to the M-CHAT-R standard. The users provide "Yes" or "No" answers regarding children's behavior, such as do they react to a gesture or respond when called?

2.2.3 Automated Scoring Algorithm of Risk Level

The system will be able to provide risk levels automatically, making sure that each child undergoes an evaluation process that is not only consistent but also standardized and objective in nature. For those children with results falling within the Low Risk category, which would include any score of 0-2, there would not be a need to pursue any additional actions unless there are any future worries. Those with scores ranging from 3-7 are considered to have Moderate Risk, automatically prompting notification about the requirement of taking some actions to further examine their behavior. In instances where the scores range from 8-20, the result will be classified as a High Risk automatically, skipping other steps to indicate the urgent requirement of referral of the client for professional diagnostic testing.

2.3 Support, Analytics and Specialized Management

In addition to the diagnosis, this system features modules which enable monitoring and assistance on a continuous basis.

2.3.1 Automated Notification System

To resolve the issue of "delayed referrals" in the conventional approach, this module helps generate automated alerts regarding upcoming screenings or follow-up consultations.

2.3.2 Reporting and Analytics

Data analytics provide an opportunity to understand how the development of a child evolves throughout time, as visualized by parents or NGOs.

2.3.3 Behavior Monitoring

This module monitors the emotional state and mood of the child, enabling caregivers to identify any possible triggers for certain behaviors. These are some of the functions of this module:

- Behavior Log Processor

  A real-time function that enables the caregiver to log emotional status, particular behavioral instances, and the corresponding environmental antecedents.

- Trigger Analyzer

  A pattern-matching algorithm that scans historical logs to identify correlations between environmental factors that repeat themselves and behavioral disruptions that occur again.

- Mood Trend Evaluator

  A computation function that calculates and charts baseline emotional movements over specified timelines, providing parents and NGOs with empirical evidence of structural improvement.

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2.3.1 Appointment Scheduling and Referral Management

This module allows parents to book appointments directly with specialist and referral centres, as well as with education centres, through the system. When a child's behavioural log or screening result shows a concerning trend, parents can make an appointment without needing to rely on a phone call, resolving the ambiguity and delay parents currently face once the limited monthly clinical sessions come to an end. The module checks real-time availability of the specialist or referral centre, records the appointment slot, and automatically issues a referral letter and appointment reminder to the relevant party. This directly replaces the current appointment method and the phone call and manual diary check process, which currently takes 1 to 3 working days and produces no automatic reminder.

2.3.2 Educational Placement and Special Education Management

Through the 3.3.2 module, parents are allowed to search, filter, and view the physical locations of specialized education and early intervention centres within their vicinity. When a child’s screening result or behavioral logs indicate a need for early learning support, parents can quickly locate nearby facilities, view available services, and access contact information without any manual search delays. This bridges the gap between clinical diagnosis and educational placement by providing a centralized directory of specialized schools.

2.3.3 Community Groups and Peer Support

An interactive communication hub provides a dedicated discussion platform where parents and caregivers can connect, share experiences, and offer mutual peer support. Users can join localized or topic-specific forums to exchange and navigate educational systems. By centralizing communication within a secure environment, system users can reduce the isolation felt by caregivers and eliminate the scattered nature of relying on external social media channels.

3.0 Project Scopes

The suggested Autism Early Screening Support System aims at updating, digitizing and centralizing the tracking process of home care and consultations for parents dealing with early signs of autism and development delay in their children. In this regard, the existing paper-based process is based mainly on disconnected paper documentation and disconnected physical flashcards which leads to confusion regarding the evaluation, increased cognitive load for the caregiver, and lack of continuity after treatment. In order to solve this problem, the suggested system provides a digital solution designed to facilitate early data collection, automated trends analysis, and seamless connection with screening services.

The architecture of the proposed system optimizes user autonomy, early identification, and institutional synchronization through five specialized core functions:

- **Behaviour Monitoring:** To ensure continuous tracking and databased insights following clinical intervention, this module provides parents with a structured framework to log routine behaviors and critical incidents. By translating daily observation data into clinical insights, the module helps identify early developmental regressions or emergent behavioral trends.

  The monitoring functionality is split between two main logging processes:

  1. **Behavior Observation Logging:** Parents log routine clinical parameters in four major areas of development: Social Interaction, Communication, Sensory Sensitivity and Repetitive and Restricted Behaviors. The information is summarized to produce the Behavioral Pattern Report, which focuses on common triggers and irregularities for each week period. All the structured logs can be saved, analyzed retrospectively and exported as a clinical PDF report.

  2. **Meltdown & Tantrum Incident Logging:** Designed specifically for severe behavioral occurrences, this process lets parents log acute behavioral episodes in a structured manner. The logging process collects relevant contextual information such as date and time, length of episode, triggers (for example, sensory over-stimulation or change in schedule), levels of intensity, and precise reaction. For greater accuracy of data, parents may include relevant media files (pictures/video) as objective clinical evidence along with timelines.

- **Center Booking Appointment:** In order to fill the post-discharge gap, the module features a real-time scheduling engine. In case the system detects any alarming behavioral patterns or serious meltdowns, parents can immediately bypass the paperwork and schedule consultation sessions directly with centers that have been verified.

  For flexible booking depending on urgency levels, the module offers two separate booking channels:

1. **Pre-assigned Specialist Model:** Suitable for therapeutic continuity where parents can book sessions with specific and dedicated specialists according to long-term treatment plans.

2. **Dynamic Assignment on Arrival:** Suitable for quick triaging, first-time visits, or parents who cannot decide on the particular specialists. In this method, families can schedule the appointment slot first. On arriving at the center, the professional team assesses the needs of the child and assigns them to the appropriate specialist.

- **Help Guideline:** In order to maintain a constant stream of user onboarding and technical self-sufficiency, this module utilizes the use of a centralized support knowledge base and triage hub. Created to minimize administrative chokepoints, the Help Center gives parents the ability to troubleshoot problems directly, through instructional resources, and communication avenues.

  This support model has been constructed through three key touchpoints:

  1. **Contact Support Gateways:** Provides multiple layers of escalation options for immediate and non-immediate help. These include an instant messaging tool showing available agents, an email address for support that receives replies within 24 hours, and a toll-free phone number.

  2. **Quick Guides:** A selected archive of procedural guides that focus on important user processes. This is where parents will be guided to conduct M-CHAT-R screeners, fill out organizational profiles (like the BPPOKU Bahagian A), understand risk classifications, and set up the behavior tracking modules.

  3. **Frequently Asked Questions (FAQs):** An expandable accordion-based FAQ engine that provides answers to commonly asked questions about platform processes, screening frequency, admin permissions, and strong privacy controls for users.

- **Special Educational Center:** The current module consists of a curated and verified regional directory of specialized autism treatment facilities and schools. For facilitating localized placements, this module acts as an intelligent directory for nearby specialized facilities. The module uses the user's geolocation to assist parents in evaluating and filtering out appropriate special education systems.

  The intelligent directory is structured into two pathways of institutions based on different family requirements:

  1. **Private Centres:** Features private institutions near the user with their unique special programs and flexibility in schedule. Each card consists of relevant information, including distance parameters, user reviews, contact registry, and badges indicating their specific therapy (e.g., Autism Therapy, Speech Therapy).

2. **Public Centres:** Consists of government-sponsored and highly subsidized programs like schools integration programs (SMK, SJK). This helps lower income families to locate their nearby public facilities without spending much money.

- **Community Groups Module:** In order to counter caregiver loneliness and build peer camaraderie, this module acts as a local community that links up caregivers to their respective communities. Users are able to browse through a range of communities from various categories, such as Diagnosis Support, Sensory Help, School/Education, and Local Get-Togethers. The community cards keep track of active member databases and daily posting stats, where parents can immediately join these communities and keep up to date with real time discussions and peer chat windows.

4.0 Project Requirements

4.1 Process 1.0

Process Specifications: 1.0 Authentication and Account Management	
Process ID	1.0
Process Name	Authentication and Account Management
Process Description	Manages secure login, logout, account registration, and password management for all system users, which are the Parents and NGO Administrators.
Input Data Flows	Login Credentials (Username/Email, Password) Registration Data (Name, Role, Contact Details) OTP Code New Password
Output Data Flows	Authenticated Session Token Account Record stored in D1 User Account Store Authentication Error Message OTP Request to Parent/Caregiver Password Reset Confirmation
Type of Process	Online
Sub-processes/ Function	1.1 Validate Login Credentials 1.2 Register New Account 1.3 Reset Password 1.4 Manage User Accounts
Process Logic	1.1 Receive login credentials from the user. 1.2 Validate credentials against D1 User Account Store. IF credentials match THEN GENERATE a session token ROUTE the user to the appropriate dashboard ELSE IF credential do not match THEN RETURN error message INCREMENT failed attempt counter ENDIF IF failed attempts is greater or equals to 3 THEN TEMPORARILY lock account SEND alert to registered email ENDIF 1.3 For new registration,validate all required fields and check

	for duplicate email in D1. IF no duplicate THEN ENCRYPT password WRITE new account record to D1 SEND welcome notification. ELSE IF duplicate found REJECT registration RETURN error ENDIF 1.4 For password reset, send OTP to registered phone or email, validate OTP, accept new password and update D1.
Logic Method Reference	Structured English
Unresolved Issues	Lockout duration after 3 failed attempts is not yet defined. OTP delivery channel and validity period have not been finalized.

4.2 Process 2.0

Process Specifications: 2.0 BPPOKU Profile Management	
Process ID	2.0
Process Name	BPPOKU Profile Management
Process Description	Digitalizes Bahagian A of the BPPOKU form by collecting and storing comprehensive child and parent or caregiver profile data across four sub-sections, which are the personal details, education details, employment details and next-of-kin details.
Input Data Flows	Personal Details (Name, MyKad, Date of Birth, Address, Contact) Education Details (Academic Level, School Name) Employment Details (Occupation, Employer, Income) Next-of-Kin (Name, Relationship, Contact)
Output Data Flows	Saved Child Profile Record in D2 Child Profile Store Profile Completion Confirmation to Parent Profile data to Process 3.0 and Process 6.0
Type of Process	Online
Sub-processes/ Function	2.1 Enter Personal Details

	2.2 Enter Education Details 2.3 Enter Employment Details 2.4 Enter Next-of-Kin Details 2.5 Confirm and Save Profile
Process Logic	2.1 Retrieve existing profile from D2 and pre-populate form fields. 2.2 Validate required fields for each sub-section. IF mandatory fields are empty THEN FLAG missing fields BLOCK save DISPLAY validation message ENDIF 2.3 Validate MyKad format. IF format invalid RETURN field-level error, ENDIF 2.4 On valid submission, write or update profile record in D2. 2.5 Return save confirmation to the Parent dashboard. 2.6 Make child age available to Process 3.0 for eligibility check.
Logic Method Reference	Structured English
Unresolved Issues	Validation rule for MyKad format needs to be confirmed against the 12 digit MyKad numbering standard. It is unclear whether parents can save a partially completed profile or must complete all four sub-sections before the record is written to D2.

4.3 Process 3.0

Process Specifications: 3.0 M-CHAT-R Screening	
Process ID	3.0
Process Name	M-CHAT-R Screening
Process Description	Delivers the 20-item M-CHAT-R checklist to eligible children, captures parent responses, automatically computes and classifies the risk, and generates follow-up actions and a screening record.

Input Data Flows	Children Details Initiate Screening Yes/No Responses (20 Items)
Output Data Flows	Eligibility Status M-CHAT-R Response Risk Level Alert to Parent/Caregiver Follow-up Actions to Parent/Caregiver Screening Record to D3 Screening Results Store
Type of Process	Online
Sub-processes/ Function	3.1 Validate Screening Eligibility 3.2 Administer 20 Question Checklist 3.3 Compute Risk Score 3.4 Determine Risk Level 3.5 Generate Follow-Up Action 3.6 Generate Screening Record
Process Logic	3.1 Retrieve child's age from D2 Child Profile Store and validate screening eligibility. IF not eligible THEN RETURN ineligibility message ENDIF 3.2 Administer the 20 question checklist and record each Yes/ No response. 3.3 Compute the risk score by tallying "At Risk" responses according to the M-CHAT-R scoring point. 3.4 Determine risk level. IF score 0-2 THEN Risk = Low ELSE IF score 3-7 THEN Risk = Moderate ELSE IF score 8-20 THEN Risk = High ENDIF 3.5 Generate follow-up action and risk level alert based on the determined risk level. 3.6 Write the final screening record to the D3 Screening Results Store
Logic Method Reference	Structured English Decision Table (Risk Level Classification)
Unresolved Issues	Exact eligible age range for the M-CHAT-R needs to be confirmed and validated against the official M-CHAT-R clinical guideline. The minimum re-screening interval before a

	parent may retake the checklist for the same child has not been defined.

4.4 Process 4.0

Process Specifications: 4.0 Notification and Reminder Management	
Process ID	4.0
Process Name	Notification and Reminder Management
Process Description	Generates and sends automated notifications and reminders to Parents based on system triggered events such as risk results, behaviour severity flags, and appointment confirmations.
Input Data Flows	Risk Level and Referral Flag Severity Flag Appointment Confirmation Recipient Contact Details Retry Flag
Output Data Flows	In-App Reminder to Parent/Caregiver Referral Notification to Specialist/ Referral Center Notification Record to D6 Notification Log Dispatch Status
Type of Process	Online
Sub-processes/ Function	4.1 Receive Trigger Event 4.2 Select Notification Template 4.3 Dispatch Notification 4.4 Log Notification
Process Logic	4.1 Receive trigger event and identify event type. 4.2 Select the appropriate notification template and priority level based on the event type. 4.3 Retrieve recipient contact details from D1 and dispatch the notification message. IF dispatch fails THEN SET retry flag TRIGGER re-dispatch ENDIF 4.4 Log the notification record to D6 Notification Log

	4.5 Retrieve existing scheduled reminders from D6 to avoid duplicate or conflicting alerts.
Logic Method Reference	Structured English
Unresolved Issues	Maximum number of retry attempts and the retry interval after a failed dispatch have not been specified. It is unclear whether parents can fully disable critical alerts or only adjust frequency for non-critical reminders.

4.5 Process 5.0

Process Specifications: 5.0 Behaviour Monitoring	
Process ID	5.0
Process Name	Behaviour Monitoring
Process Description	Captures and logs routine behaviour observations and acute incident reports, analyzes trigger patterns and mood trends, and generates summarized observation history for parents.
Input Data Flows	Daily Observation Log Incident Log Historical Log
Output Data Flows	Timestamped Entries to D4 Behaviour Log Store Trigger Correlations Mood Ratings Summary Report Observation History to Parent/ Caregiver
Type of Process	Online
Sub-processes/ Function	5.1 Log Behaviour Observation 5.2 Log Incident Report 5.3 Analyze Trigger Pattern 5.4 Compute Mood Trend 5.5 Generate Pattern Summary 5.6 Output Observation History
Process Logic	5.1 Receive and timestamp daily behaviour observation entries, write to D4. 5.2 Receive and timestamp incident report entries, write to

	D4. 5.3 Analyze trigger patterns across historical logs retrieved from D4. 5.4 Compute mood trends based on logged mood ratings over the selected period. 5.5 Generate pattern summary combining trigger correlations and trend data. IF a behaviour repeats 2 or more times within the period THEN FLAG as “Repeated Pattern” ENDIF 5.6 Output the observation history and summary data to the Parent/ Caregiver.
Logic Method Reference	Structured English
Unresolved Issues	The exact threshold and time window used to qualify a behaviour as a “Repeated Pattern” have not been finalized. Maximum file size/ format and storage handling for uploaded video or image evidence still need to be determined.

4.6 Process 6.0

Process Specifications: 6.0 Reporting and Analytics	
Process ID	6.0
Process Name	Reporting and Analytics
Process Description	Aggregates screening, behaviour, and appointment data into formatted reports and a unified analytics dashboard for parents, specialists, and NGO administrators.
Input Data Flows	Screening Results from D3 Screening Results Store Behaviour Logs from D4 Behaviour Log Store Appointment History from D5 Appointment Store
Output Data Flows	Formatted M-CHAT-R Reports Formatted Behaviour Reports Aggregated Analytics Reports to NGO Administrator Unified Dashboard Shared Reports to Specialist/ Referral Center Report Output to Parent/ Caregiver

Type of Process	Online
Sub-processes/ Function	6.1 Retrieve Screening Data 6.2 Retrieve Behaviour Data 6.3 Retrieve Appointment Data 6.4 Generate M-CHAT-R Reports 6.5 Generate Behaviour Reports 6.6 Compile Analytics Dashboard 6.7 Export and Share Report
Process Logic	6.1 Retrieve screening data and score trends from D3. 6.2 Retrieve behaviour data and observation summaries from D4. 6.3 Retrieve appointment history and attendance data from D5. 6.4 Generate M-CHAT-R report from score trends. 6.5 Generate behaviour report from observation summaries. 6.6 Compile all formatted reports into the unified analytics dashboard. 6.7 On request, export and share the report. IF recipient equals to NGO Administrator THEN SEND aggregated analytics report ELSE IF recipient equals to Parent THEN ENABLE print/ export ELSE IF recipient equals to Specialist THEN SHARE report ENDIF
Logic Method Reference	Structured English
Unresolved Issues	The frequency of analytics dashboard refreshes has not been decided. The exact permission boundaries for NGO Administrators viewing aggregated and individual child level data still need to be confirmed for privacy compliance.

4.7 Process 7.0

Process Specifications: 7.0 Appointment and Centre Management	
Process ID	7.0
Process Name	Appointment and Centre Management

Process Description	Handles the full cycle of appointment booking, viewing, rescheduling, and cancellation while including centre and specialist search.
Input Data Flows	Filter Preferences Selected Centre and Specialist, Date/ Time Slot Appointment ID, Cancellation/ Reschedule Request Rating and Comment, Specialist/ Centre ID Diagnostic Feedback, Consultation Details
Output Data Flows	Filtered Centre and Specialist List Book Confirmation Appointment Confirmation Alert Updated Appointment Status Alert New Appointment Record to D5 Appointment Store Review Record to D5 Appointment Store
Type of Process	Online
Sub-processes/ Function	7.1 Search Centres and Specialist 7.2 Select and Book Appointment 7.3 Cancel or Reschedule 7.4 Submit Review 7.5 Receive Consultation Details
Process Logic	7.1 Search centres and specialists from D5 based on filter preferences and return filtered list. 7.2 Select and book appointment. IF selected slot is available THEN WRITE new appointment record to D5 RETURN booking confirmation ELSE RETURN slot unavailable error ENDIF 7.3 For reschedule/ cancellation requests, validate appointment ID against D5. IF valid THEN UPDATE appointment status RETURN updated status alert ENDIF 7.4 On consultation completion, receive rating/ comment and submit review, write review record to D5. 7.5 Receive consultation details and diagnostic feedback from Specialist/ Referral Center for the corresponding appointment.

Logic Method Reference	Structured English
Unresolved Issues	Cancellation/ reschedule policy has not been defined. It is unclear whether reviews are moderated before being published or visible to other parents immediately after submission.